1/24/19

Lecture 4.

Mass flux/area

1-D continuity.

$$\dot{m} = \rho V A = \rho_0 V_0 A_0$$

or $$\frac{\dot{m}}{A_0} = \rho_0 V_0$$

Know that thrust scales w/ $\dot{m}$

$$T \sim \dot{m}(V_e - V_0)$$

From dyn. pr. $$q_0 = \frac{1}{2}\rho_0 V_0^2$$

$$\Rightarrow \quad \rho_0 V_0 = \frac{2q_0}{V_0}$$

$$\frac{\dot{m}}{A} = \rho_0 V_0 = \frac{2q_0}{a_0 M_0}$$

Note: often want to maintain const $\frac{\dot{m}}{A}$ to provide sufficient $\dot{m}$ into engine. Can see from above that if we were to maintain $\rho_0 V_0$ & const q_0, we can only operate @ a single M #, which is difficult. If we maintain, $q_0 = \text{¢}$ for structural reasons, then $\frac{\dot{m}}{A}$ varies inversely w/ M_0

$\Rightarrow$ To keep the thrust constant $\Rightarrow$ keep $\dot{m}$ const

$\Rightarrow$ increase the capture area $\Rightarrow$ use forebody compression where @ design cond^n the forebody shock hits the cowl lip.

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Hence, we will see that hypersonic vehicles use the entire underside of the vehicle as a part of the inlet capture to greatly increase A_0.

Boundary layers in hypersonic vehicles

- B.L. can be laminar, turbulent, or transitional
- Laminar can be good due to lower skin friction & heat transfer. However, they are also prone to separation (w/ adverse $\frac{dp}{dx}$) & shock. Tend to be thicker.
- Use Re # to determine the nature of B.L.

$$Re \equiv \frac{\text{Inertial forces}}{\text{Viscous forces}} = \frac{\rho V L}{\mu} = \frac{V L}{\nu}$$

Define a freestream Re

$$Re_0 = \frac{V_0 L}{\nu_0}$$

L: characteristic dimension of vehicle

- In gen. when $Re < Re_{crit} \rightarrow$ laminar b.l.

  $Re \gg Re_{crit} \rightarrow$ turb.

  $Re \approx Re_{crit} \rightarrow$ transitional.

- Determination of Re_{crit} is not trivial $\because$ $Re_{crit} = f(M_0, \frac{dp}{dx}, u'_{rms}, \text{3D effects}, \text{rxns})$...
  - Generally, for vel. w/ $M \leq 10$, can assume fully turb. b.l. if $Re > 10^7$

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Eg: Assume vehicle of $L = 30\,m$ @ $M = 4.0$, $H = 25\,km$. What is q_0 & the nature of B.L.?

$H = 25\,km$

$$\Rightarrow \frac{p}{p_{SL}} = 0.02516 \; ; \quad \frac{\rho}{\rho_{SL}} = \cancel{0.03272}\; 0.03272$$

$$q_0 = \frac{1}{2}\gamma p_0 M_0^2$$

$$= \frac{1}{2} \times 1.4 \times 0.2516 \times \underbrace{1.01\times10^3}_{p_{SL}} \times 4^2$$

$$q_0 = 28.5\,kPa.$$

$$Re_0 = \frac{V_0 L}{\nu_0} \; ; \quad V_0 = a_0 M_0$$

$$\frac{T_0}{\underbrace{T_{SL}}_{288K}} = 0.7689 \Rightarrow T_0 = 221\,K. \; ; \; V_0 = \sqrt{\gamma R T_0}\, M_0 = 1200\,m/s$$

$$\mu_0 (25km) = 1.46e-6 \left[\frac{T_0^{3/2}}{T_0 + 111}\right] = 14.4\times10^{-6}$$

$$Re_0 = \frac{1200 \times 30 \times \rho_0}{\mu_0}$$

$$\rho_0 = 0.03272 \times \underbrace{\rho_{SL}}_{1.225} = 40\times10^{-3}\,kg/m^3$$

$$Re_0 = 100e6 \Rightarrow \text{Turb.}$$

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What about $M = 10$ & same q_0 ?

$$q_0 = 28.5 \text{ kPa}.$$

$$q_0 = \frac{1}{2}\gamma p_0 M_0^2$$

$$p_0 = 0.407 \text{ kPa}.$$

$$\Rightarrow \frac{p_0}{p_{SL}} = \frac{0.4}{101} = 0.4 \times 10^{-2}$$

$$h = 37.5 \text{ km}$$

$$\frac{\rho_0}{\rho_{SL}} = 0.509 \times 10^{-2} \; ; \; \frac{T_0}{T_{SL}} = 0.84 \Rightarrow T_0 = 250 \text{ K}.$$

$$\rho_0 = 6 \times 10^{-3} \text{ kg/m}^3$$

$$\boxed{Re_0 \approx 56e6}$$

$\Rightarrow$ B.L. may now be in transition.

$\rightarrow$ In general, the faster we go, the higher we must fly, the lower the Re.

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What about our assumption regarding continuum fluid mechanics @ these high altitudes?

Look @ knudsen #

$$Kn \equiv \frac{\text{mean free path}}{\text{char. length}} = \frac{\lambda}{L}$$

Look @ 4 regimes

$Kn < 0.01 \Rightarrow$ (continuum regime)

$0.01 < Kn < 0.1 \Rightarrow$ slip flow

$0.1 < Kn < 3 \rightarrow$ transition regime

$Kn > 3 \rightarrow$ free molecular flow (spacecrafts)

λ scales w/ ~~inverse~~ inverse of $\rho \Rightarrow \lambda \uparrow$ w/ alt.

It can be shown that $Kn \approx 1.26\sqrt{\gamma}\left(\frac{M_\infty}{Re}\right)$

So for e.g. @ $L = 30m$, $M_\infty = 10$, $q_\infty = 28.5 kPa$

$$Kn = 1.26 \times (1.4)^{1/2}\left(\frac{10}{28e6}\right) = 540 \times 10^{-9}.$$

$\Rightarrow$ even @ very high altitudes, well within continuum regime

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Composition of air

Normally assume air : 0.21 O_2 & 0.79 N_2. (molefraction)

- This is a good assumption until we get to temps high enough that dissociation becomes important

$$N_2 \rightleftharpoons 2N \; ; \qquad O_2 \rightleftharpoons 2O$$

- Chemical eq^m based on the ~~reaction~~ minimization of Gibbs free energy.
- Equibm const ~~is~~ (K_p) is a very strong f(T) & it may or may not be a funct^n of pr.
- We can express the eq^m composition in terms of the eq^m const K_p; K_p will only be a funct^n of temp, but a very non-linear one.

$$K_p(T) = \prod p_i^{(\nu_i'' - \nu_i')}$$

→ p_i : partial pr. of species 'i'
ν_i' = stoic. coeff of reactants
ν_i'' = stoic coeff of products.

- Dalton's Law : $p_i = \frac{N_i}{N} p$; N_i = # of moles of i
N : total # of moles $= \sum_i N_i$
p : total p $\Rightarrow p = \sum_i p_i$

$$\sum_i \frac{N_i}{N} = 1$$

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Using ideal gas law.

$$\boxed{K_p(T) = \prod \left[N_i^{(\nu_i'' - \nu_i')} \right] \left[\frac{p}{N} \right]^{\Delta \nu_i}}$$

$$\rightarrow \Delta \nu_i = \sum_i (\nu_i'' - \nu_i)$$

p: total pr.

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